UNIT – I

Introduction to Information Retrieval Systems: Definition of Information Retrieval System, Objectives of Information Retrieval Systems, Functional Overview, Relationship to Database Management Systems, Digital Libraries and Data Warehouses
Information Retrieval System Capabilities: Search Capabilities, Browse Capabilities, Miscellaneous Capabilities

Short Questions

1. Why is it necessary to retrieve an information?
2. What is CWP?
3. Draw the thesaurus for the term COMPUTER.
4. What is Browse capability?
5. What is a search engine?
6. Explain recall.
7. What is zoning?
8. Define the term Masking.
9. What do you mean by precision?
10. What is the primary goal of an information retrieval system?
11. What is Parsing?
12. Discuss about ranking.

Long Questions

1. Define Information Retrieval System. Explain the objectives of the Information Retrieval System.
2. Write a brief note on Information Retrieval System capabilities.
3. How the Information Retrieval System is related to Database Management System?
4. What problems need to address when using a DBMS as part of an Information retrieval System?
5. What is the difference between the concept of a "Digital Library" and an Information Retrieval System?
6. Classify two standards of IRS capabilities.
7. Describe functional overview of information retrieval systems (IRS)?
8. What are the various ways of obtaining the information?
9. How the information system is evaluated and what are the measures used in it.
10. How digital libraries and data warehouses are housed as a source of Information systems.
11. What is the impact on precision and recall in the use of Stop Lists and Stop Algorithms?
12. What are the similarities and differences between fuzzy searches and term masking? Mention the potential of each.
13. How Selective dissemination of information search is done to the user?
14. Discuss the objectives of information retrieval systems.
15. Summarize about the miscellaneous capabilities.
16. Explain how Thesaurus are used to expand a query.

17. Discuss the difficulties of a user being able to correlate his search to the Hit file. What approach would you use to overcome these problems.
18. Briefly explain text normalization process.
19. Discuss about miscellaneous capabilities.

UNIT - II

Cataloging and Indexing: History and Objectives of Indexing, Indexing Process, Automatic Indexing, Information Extraction
Data Structure: Introduction to Data Structure, Stemming Algorithms, Inverted File Structure, N-Gram Data Structures, PAT Data Structure, Signature File Structure, Hypertext and XML Data Structures, Hidden Markov Models.

Short Questions

1. What is proximity? How can it be calculated?
2. What is the importance of thesauri?
3. What is Information Extraction?
4. Define Hypertext Structure.
5. Define stop words and their significance in text processing.
6. What is the meaning of *,?, $?
7. What are stop-words?
8. Define Stemming.
9. Discuss about hypertext history
10. Explain precoordination and linkages.
11. What are Hypertext Linkages?
12. What are the advantages of Signature file structure?
13. Explain the use of XML.
14. N-gram data structures. Describe how they affect precision and recall.
15. What is indexing and its objectives?

Long Questions

1. Illustrate porter stemming algorithm with an example.
2. Illustrate how stemming and lemmatization improve retrieval accuracy
3. Briefly explain how Automatic Indexing, Information Extraction works.
4. Explain in detail about “Inverted File Structure”.
5. Explain about classes of Automatic Indexing in detail.
6. Define stemming and What is the advantage of Stemming algorithm?
7. Describe the similarities and differences between Term Stemming algorithms
8. Discuss about Bayesian Model.
9. Differentiate the process of information extraction from the process of document indexing?

10. Briefly explain text normalization process.
11. Discuss about miscellaneous capabilities. Illustrate porter stemming algorithm with an example.
12. Explain the Indexing process in detail.
13. What is Hypertext Data Structure? Where is this data structure extensively used in? Explain.
14. Compare indexing by term and indexing by concept.
15. How does the process of information extraction differ from the process of document indexing?
16. Write about PAT data structures.
17. What are the advantages of N-Gram data structure? Write the advantages of N-grams. How N-grams is used in spelling error detection and correction?
18. Explain about Ranking algorithm with respect to Hidden Markov Model.

UNIT – III

Automatic Indexing: Classes of Automatic Indexing, Statistical Indexing, Natural Language, Concept Indexing, Hypertext Linkages
Document and Term Clustering: Introduction to Clustering, Thesaurus Generation, Item Clustering, Hierarchy of Clusters

Short Questions

1. What is indexing?
2. With the help of an example, explain the concept of statistical indexing.
3. What is Multimedia indexing?
4. Elaborate complete term relation method.
5. Define clustering.
6. What is Hierarchy Clustering?
7. What is the process involved in Term clustering?
8. Define Natural Language Processing.
9. Describe statistical indexing.
10. Define dendogram.
11. What is a cluster? Give the hierarchy of Clusters.
12. Name two types of retrieval models commonly used.
13. Provide two examples illustrating how indexing enhances information accessibility?

Long Questions

1. What is thesaurus generation and how do you generate?
2. List any two key characteristics of intelligent agents.
3. Write a short note on clustering.
4. Explain concept indexing in detail.
5. Describe the term weighting schemes in vector space models.

6. Compare probabilistic and automatic indexing model with examples Critically evaluate different methods for handling synonymy in text retrieval.
7. What is automatic indexing and various classes of automatic indexing?
8. Define Information Retrieval System. What is item clustering?
9. Explain the concept of term frequency in vector space models.
10. Which one is efficient? Boolean query or Natural Language Query. Explain
11. Compare and contrast manual clustering and Automatic Term Clustering.
12. Clearly bring out the steps of the process of clustering.
13. Explain Manual clustering.
14. Explain One Pass Assignments.
15. Summarize Concept Indexing.

UNIT – IV

User Search Techniques: Search Statements and Binding, Similarity Measures and Ranking, Relevance Feedback, Selective Dissemination of Information Search, Weighted Searches of Boolean Systems, Searching the INTERNET and Hypertext
Information Visualization: Introduction to Information Visualization, Cognition and Perception, Information Visualization Technologies

Short Questions

1. What is a search statement?
2. What is Visualization?
3. What is relevance feedback?
4. Define cognition with example.
5. Explain query binding.
6. What is perception with example?
7. What is positive and negative feedback?

Long Questions

1. What are the weighted searches of boolean systems?
2. List some areas where information visualization and presentation can help the user.
3. Define Binding.
4. Why is relevance feed required in User Search Techniques? Why is Search Statements and binding required?
5. Mention two factors that influence the ranking of search results.
6. Why is relevance feedback required in User Search Techniques? Explain?
7. Write short notes on the following with examples i) Similarity measures ii) Ranking algorithms

8. With an example, describe in detail about relevance feedback and its impact and effect of relevance feedback.

UNIT – V

Text Search Algorithms: Introduction to Text Search Techniques, Software Text Search Algorithms, Hardware Text Search Systems
Multimedia Information Retrieval: Spoken Language Audio Retrieval, Non-Speech Audio Retrieval, Graph Retrieval, Imagery Retrieval, Video Retrieval

Short Questions

1. What is text search algorithm?
2. Define Multimedia information retrieval.
3. List three applications for content-based video.
4. What are the approaches to the data stream.
5. Define tracking and detection.
6. Explain Tree structure.
7. What is a cone-tree.
8. Explain city scape.

Long Questions

1. Discuss Fast Data Finder technique
2. Explain the Rabin-karp algorithm with an example.
3. What kind of features in audio can be used to index the content?
4. Explain Hardware text search systems in detail?
5. Construct a Bad Match Table for the pattern "teammast", text "welcometoteammast".
6. List three types of media beyond text that could be indexed for subsequent retrieval.
7. Discuss the future trends in multimedia information retrieval and their potential applications.
8. Describe the concept of personalization in web search.
9. What is content-based image retrieval? List two techniques used for video information retrieval.
10. Explain full text scanning
11. Classify various Software text search algorithms.
12. Classify various multimedia retrieval.
13. Describe the need for information visualization.
14. Differentiate hardware versus software text search algorithms.
15. Explain Information Visualization Technologies Illustrate Knuth-Morris-Pratt Algorithm with an example.

16. Write about information visualization technologies.
17. Summarize Video Retrieval
18. Explain Boyer-Moore Algorithm with an example.
19. Write short notes of the following: (a)Spoken Language Audio retrieval (b)Imagery retrieval
20. Discuss the evolution of information retrieval systems and their impact on modern applications.
21. Explain with examples how metadata is used to enhance retrieval effectiveness.